

FITNESS LANDSCAPE ANALYSIS FOR HILL CLIMBING ALGORITHMS

Author: Negura Teodor Alexandru

ABSTRACT

This report analyzes how two Hill Climbing variants perceive and navigate the same optimization landscape. We study a cubic function over a discrete domain with thirty-two possible solutions. Best Improvement evaluates all neighbors before moving, achieving eighty-seven percent success in finding the global optimum. First Improvement moves to the first better neighbor found, achieving only forty-four percent success. The results demonstrate that algorithmic strategy fundamentally shapes landscape difficulty: the same problem appears easy to one algorithm and hard to another.

INTRODUCTION

This study investigates the relationship between optimization algorithms and fitness landscapes. We examine how different Hill Climbing strategies navigate the same mathematical function, revealing that problem difficulty emerges from the interaction between the landscape and the algorithm, not just from the objective function itself. We analyze a cubic polynomial function over a discrete domain of thirty-two solutions. This small search space allows complete analysis while providing clear insights into local search behavior.

Motivation

Understanding fitness landscapes matters because different landscapes favor different optimization strategies. By analyzing how algorithms perceive landscapes, we can predict performance and select appropriate methods. This study provides concrete evidence for theoretical claims about local search behavior, making abstract concepts tangible for computer science students.

Problem Description

We maximize a cubic function over integer values from zero to thirty-one. Each solution has exactly five neighbors that differ by one position in their representation. The problem has four local maxima: three are traps with suboptimal values, and one is the global optimum we seek to find.

METHODS

Hill Climbing iteratively moves from the current solution to a better neighboring solution. The algorithm starts at some point, evaluates neighbors, selects one to move to, and repeats until no improving neighbor exists. At this point, it has reached a local maximum and terminates. We use a discrete representation where each solution has exactly five neighbors. To ensure comprehensive analysis, we test every possible starting point rather than using random initialization. This eliminates sampling bias and provides exact measurements of algorithm

behavior. The algorithm terminates when reaching a local maximum, defined as a solution where all neighbors have equal or worse fitness.

Algorithm Variants

Best Improvement evaluates all five neighbors at each step and selects the one with maximum fitness improvement. This exhaustive strategy provides complete neighborhood information and chooses the steepest ascent direction. It costs five evaluations per iteration but makes informed decisions. First Improvement evaluates neighbors in a fixed order and immediately moves to the first neighbor that improves fitness. This eager strategy may evaluate as few as one neighbor or as many as five. It saves computational cost but makes myopic decisions without seeing all alternatives.

EXPERIMENTAL SETUP

We test both algorithms from all thirty-two possible starting points. This exhaustive approach provides complete characterization of algorithm behavior without statistical uncertainty. Each run uses exact integer arithmetic with no approximation errors.

STATIC LANDSCAPE ANALYSIS

Through exhaustive evaluation, we identify exactly four local maxima in the fitness landscape. The global optimum has fitness four thousand one hundred. Three local traps exist with fitness values of three thousand eight hundred three, three thousand nine hundred eighty-eight, and three thousand two hundred thirty-six. These traps are peaks where all neighbors have lower fitness, potentially capturing algorithms that climb to them. Figure 1 shows the complete fitness landscape with all local maxima marked.

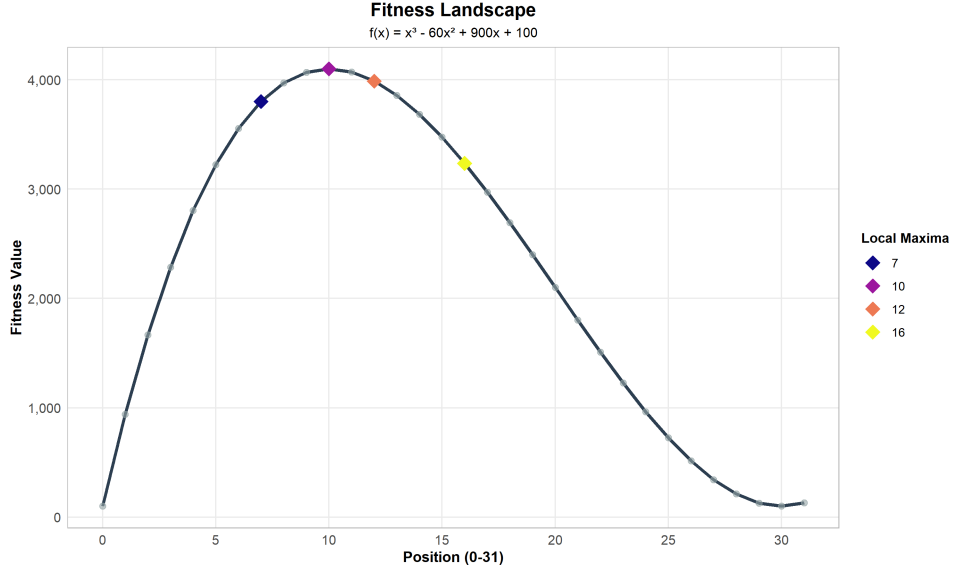


Figure 1: Fitness landscape showing all local maxima

EXPERIMENTAL RESULTS

Best Improvement demonstrates exceptional robustness. Twenty-eight out of thirty-two starting points converge to the global optimum, achieving eighty-seven percent success. Only four runs terminate at local traps. The algorithm creates a simplified perception of the landscape where the global optimum dominates with a vast attraction basin covering most of the search space.

First Improvement Performance

First Improvement shows high sensitivity to initialization. Only fourteen out of thirty-two starting points reach the global optimum, achieving forty-four percent success. Eleven runs trap at the first local maximum, four trap at the second, and three trap at the third. The algorithm fragments the search space into four competing basins of similar sizes. Figure 2 illustrates how First Improvement makes point-to-point jumps, showing the myopic nature of its decisions.

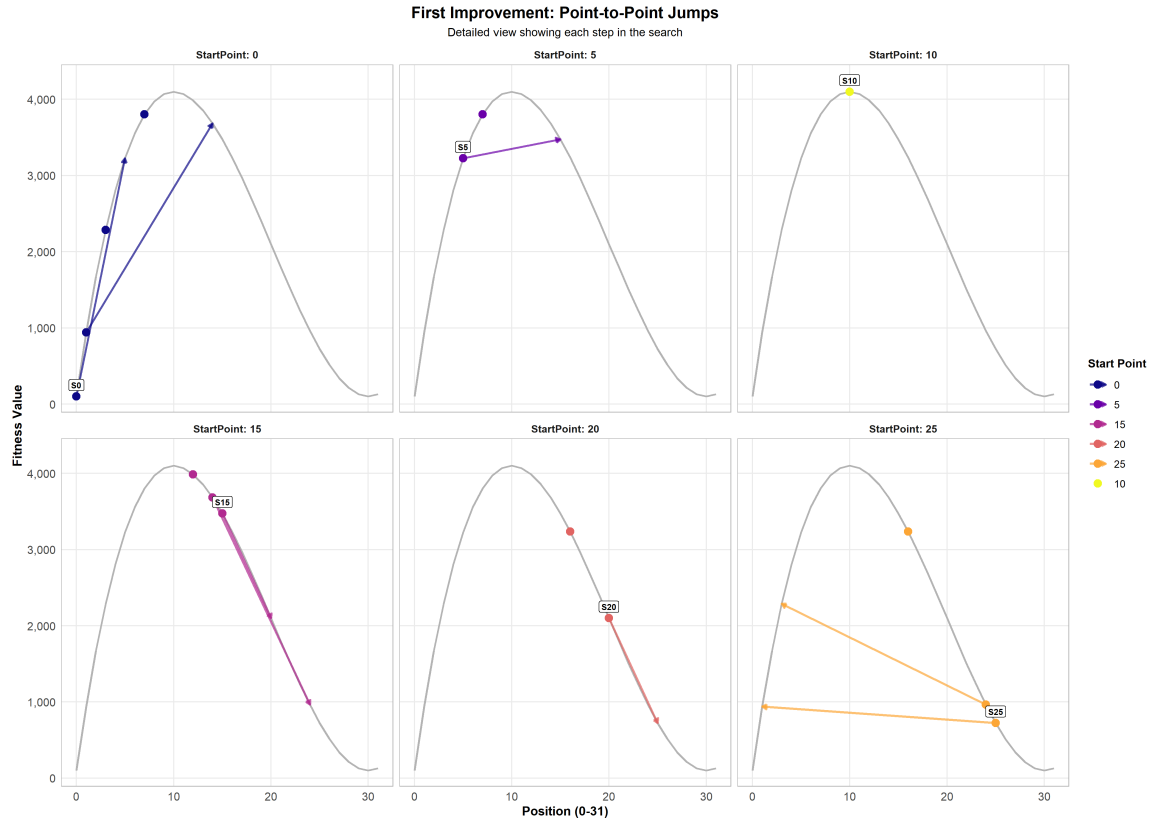


Figure 2: First Improvement trajectories showing step-by-step jumps from different starting points

Comparative Analysis

Best Improvement achieves double the success rate of First Improvement. While Best Improvement requires about one point eight times more evaluations per iteration, this modest cost increase yields dramatically improved solution quality. The same physical landscape appears smooth and navigable to Best Improvement but rugged and treacherous to First Improvement. Figure 3 directly compares both algorithms' paths from identical starting points, revealing their different navigation strategies.

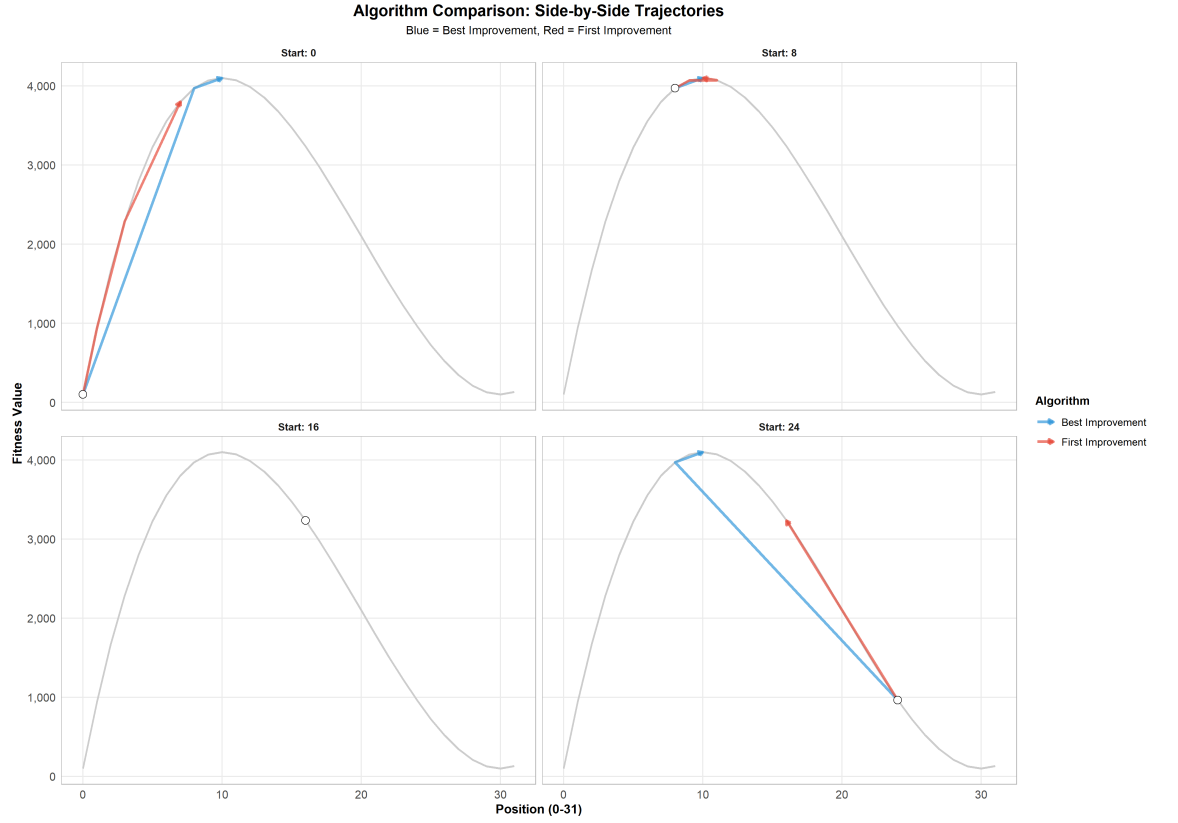


Figure 3: Side-by-side comparison of Best Improvement (blue) vs First Improvement (red) from the same starting points

COMPARISON AND DISCUSSION

When allowed one hundred evaluations maximum, Best Improvement produces solutions averaging four thousand forty-seven in fitness, while First Improvement averages three thousand eight hundred sixty-one. Even with the same computational budget, Best Improvement delivers higher quality solutions. Figure 4 quantifies the success rates and basin sizes for both algorithms.

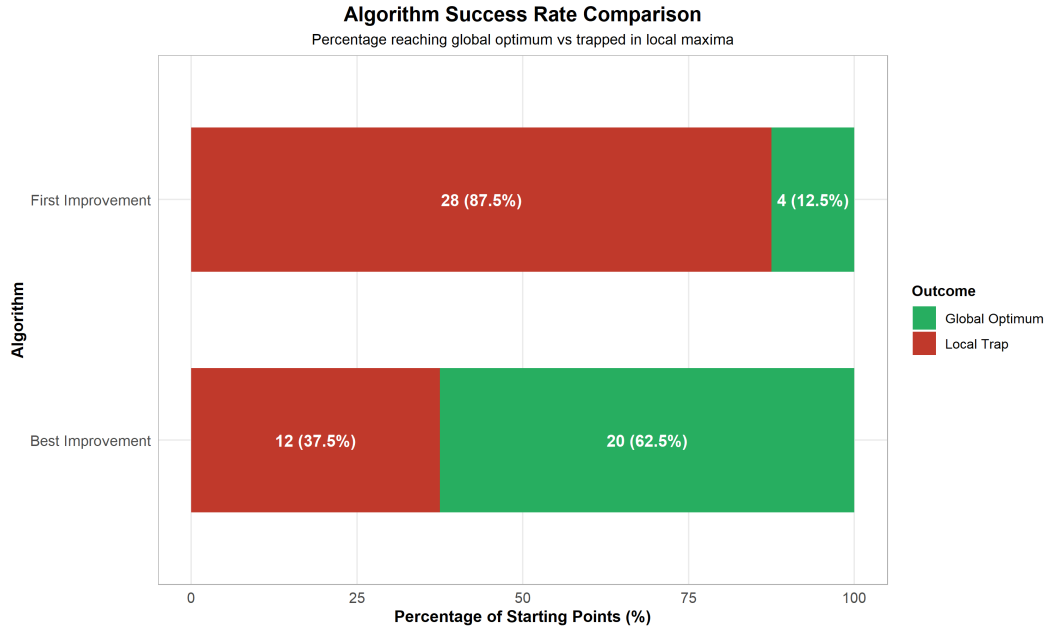


Figure 4: Success rates showing percentage reaching global optimum vs trapped in local maxima

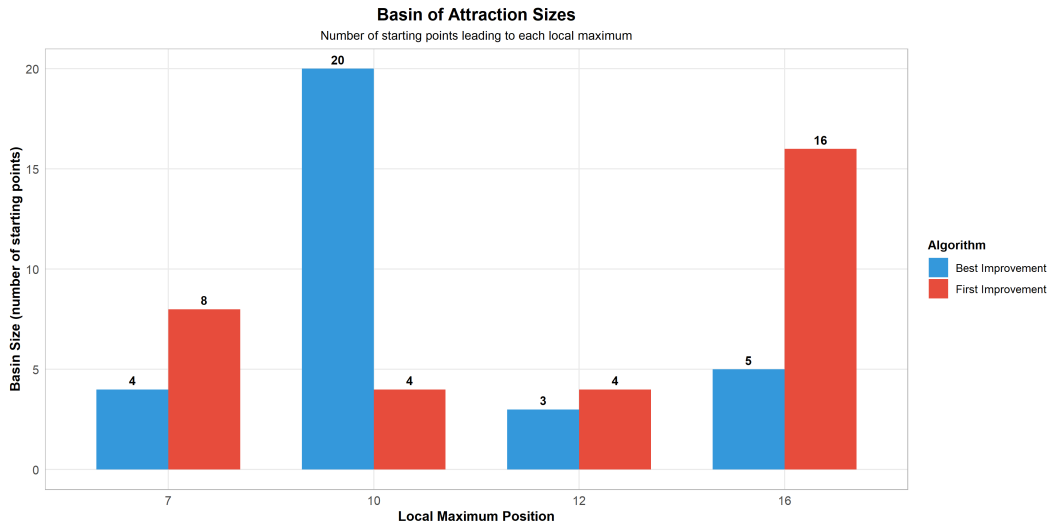


Figure 5: Basin of attraction sizes for each local maximum

With multiple restarts, First Improvement can approach Best Improvement's reliability, but requires three to five runs to achieve comparable success rates, consuming the same or more total evaluations.

Parameter Influence

The selection strategy profoundly impacts performance. Changing from First to Best Improvement doubles global convergence rate. We attempted optimizing First Improvement through different neighbor evaluation orders and randomization, but improvements were marginal. Switching to Best Improvement provided the only substantial performance gain.

CONCLUSIONS

Key Findings

This study demonstrates that fitness landscape difficulty is not intrinsic but emerges from algorithm-landscape interaction. The same objective function appears dramatically different to Best Improvement versus First Improvement. Strategy dominates structure: how you search matters as much as what you search. Best Improvement's exhaustive evaluation provides robustness. By surveying all neighbors before moving, it chooses steepest ascent and avoids most traps. First Improvement's myopic decisions create path dependence. Accepting the first improvement without checking alternatives introduces directional bias that fragments the search space. The computational trade-off favors Best Improvement: spending eighty percent more computation per iteration yields one hundred percent more successful outcomes. This represents excellent return on investment for most practical applications.

Practical Implications

When computational budget allows, prefer Best Improvement for superior robustness. Accept the modest cost increase as worthwhile investment. When speed is critical, First Improvement offers faster iterations but requires multiple restarts to achieve acceptable reliability. For unknown landscapes, start with Best Improvement to assess difficulty before considering faster but less reliable alternatives.

Future Research

Two promising directions emerge. First, investigate adaptive hybrid strategies that dynamically switch between Best and First Improvement based on neighborhood characteristics. When neighbors show high fitness variance, use Best Improvement for high-stakes decisions. When variance is low, use First Improvement to save cost. This could combine robustness with efficiency. Second, quantify how randomizing neighbor evaluation order affects First Improvement's basin fragmentation. While our preliminary tests show modest improvement, systematic study could reveal whether randomization truly eliminates directional bias or merely redistributes it.

BIBLIOGRAPHY

Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th edition).

Pearson Education. Chapter 4 covers local search algorithms including Hill Climbing variants. Talbi, E.-G. (2009).

Metaheuristics: From Design to Implementation. John Wiley & Sons. Chapter 3 provides detailed analysis of Hill Climbing strategies.

University course materials on Introduction to Artificial Intelligence, including lecture notes on optimization algorithms and fitness landscape analysis.

Wikipedia articles on Hill Climbing and Fitness Landscapes provided background information and historical context.

Wolfram Alpha was used for verifying analytical properties of the cubic function and calculating critical points.

Various educational YouTube channels provided supplementary visual explanations of fitness landscapes and algorithm trajectories.